Constructing Cloud-based Lab Infrastructure for Cyber Security Education

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Introduction

Our project focuses on constructing a cloud-based virtual laboratory for cybersecurity education. Compared with the existing virtual labs, testbeds, games, exercises, and competitions developed for cybersecurity education, our project is highly-available, highly-configurable, highly-scalable, and free, to facilitate various purposes of cybersecurity education. We have designed and are implementing our project upon *CloudLab*[1], a free infrastructure dedicated for academic usage and sponsored by the NSF. Our current implementation is available online for students and educators.

Motivation

- Most of the existing projects focus on:
  - Designing stand-alone cybersecurity-related hands-on labs.
  - Building the testbeds for education and research.
  - Training student's individual hands-on capabilities in cybersecurity.
- Little effort has been imposed on how to construct sophisticate virtual network for attacking, defensing, and management.

The Features of Our System

Our system is designed to provide the following features:

- Amplify the instructors' capabilities to create, update, backup, and restore the artifacts in a virtual network.
- Scalable in the cloud without financial and maintenance cost.
- Allow security exploits to be constrained.
- Allow the educators to control the bare-metal physical machines, as well as to fully control the hypervisor and OpenStack software.

1. *CloudLab*: https://www.cloudlab.us/

System Design

Overview

- Construct three domains: ***Attack Domain***, ***Target Domain***, and ***Control Domain***.
- Each domain can properly be separated.
- The exploits can be launched, monitored, constrained, and studied later.

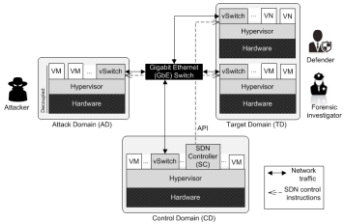

Fig. 1 The System Overview

Key Technologies

- Use **`CloudLab`** (https://www.cloudlab.us/) as the free cloud infrastructure.
- Use **`OpenStack`** (https://www.openstack.org/) to construct the virtual network.
- Use **`Python`** to profile the virtual network artifacts.

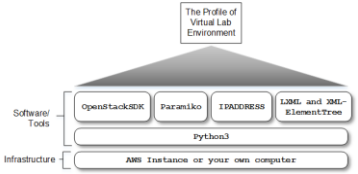

Fig. 2 The Key Technologies Used in This Project

Pervasive Adoption

- The profile of virtual environment is saved in the .XML format.
  - Easy to understand
  - Easy to adopt and customize
- The profiled network artifacts:
  - Security domains
  - VM instances
  - Network topology
  - Subnet, router, gateway, etc.
- The source code of this project can be accessed from:
  - Github
  - Bitbucket

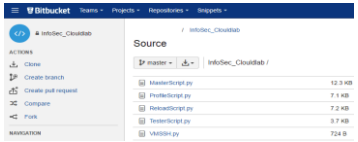

Fig. 3 The Code Repository at Bitbucket

```
<?xml version="1.0" ?>
<Profile>
  <Interfaces>
    <Interface ID="213fbc80-84c7-43e4-acc3-40de8fe434eb">
      <Interface_Name/>
      <Network_ID>5a742f66-0a82-498a-b5f1-5d02240750ec</Network_ID>
      <Interface_Admin_state_UP>True</Interface_Admin_state_UP>
      <Connected_Device_ID>2e5136f9-d4b7-432c-9a19-c882df8aade</Connected_Device_ID>
      <InterfaceSubnet ID="12a8c86f-5b67-4981-b3de-ad91189cc856">
        <InterfaceIP>128.110.155.153</InterfaceIP>
      </InterfaceSubnet>
    </Interface>
...
```

Fig. 4 The Profile of Virtual Environment in XML format

Tasks and Milestones

Current status

- Implemented the major components of the system.
- Implemented a library and a command-line interface.
- The virtual environment and VM instances can be profiled and deployed automatically.

Tab. 1 The Virtual Environment under Construction

Constructing Virtual Environment (Work in Progress)			
Sub-tasks	Required Software, Libraries, and Languages	Status	
		Prototype	Refinement
1: Setup the control domain (CD)	OpenStack Python API and AlienVault USM	✓	✓
2: Start/shutdown the VMs	OpenStack Python API	✓	✓
3: Profile the virtual environment of the labs	OpenStack Python API	✓	✓
4: Script the snapshotting and restoration of each domain	OpenStack Python API and Hypervisor-specific command and API	✓	✓
5: Change the topology of the overlay network	Open vSwitch (OVS) API	⚠	
6: Change routing rules	Open vSwitch (OVS) API	⚠	

Tab. 2 The Security-related Virtual Hands-on Labs

Security-related Virtual Labs (Work in Progress)		
Lab Description	Status	
	Prototype	Refinement
1: Cloud-based malware construction and virtual machine forensics	✓	✓
2: Fine-grained Access Control with Attribute-based Encryption (ABE)	✓	✓
3: Processing encrypted data with Homomorphic Encryption (HE)	✓	✓

Acknowledgement

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